

Approximating the Score Function using Optimal Transport

Curves and Surfaces 2026

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Laboratoire de Mathématiques d'Orsay

June 11th

(Badly and Slowly) Approximating the Score Function using Optimal Transport

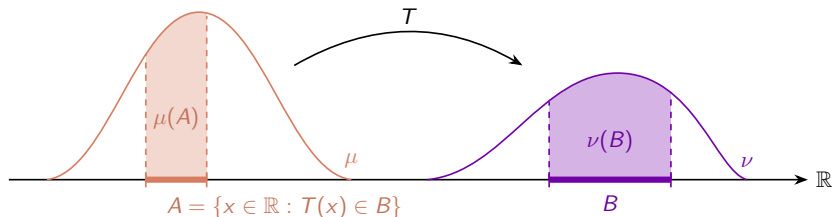
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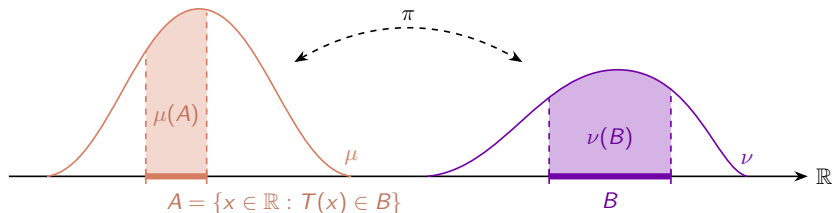
Quick reminders on Optimal Transport 1/3



- ▶ Given ground cost $c(x, y)$
- ▶ Among all maps T satisfying $\forall B, \nu(B) = \mu(T^{-1}(B))$
- ▶ Find the one minimising the average transport cost

$$\int_{\mathbb{R}^d} c(x, T(x)) \mu(dx)$$

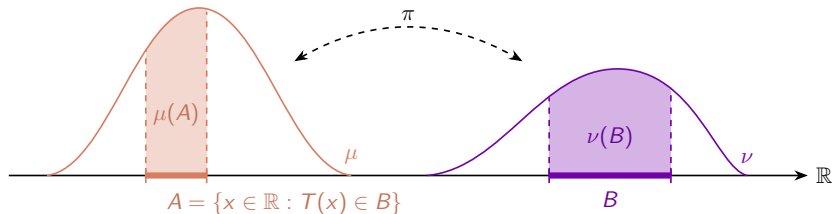
Quick reminders on Optimal Transport 1/3



- ▶ Given ground cost $c(x, y)$
- ▶ Among all maps T satisfying $\forall B, \nu(B) = \mu(T^{-1}(B))$
Among all **transport plans** π with marginals μ and ν
- ▶ Find the one minimising the average transport cost

$$\int_{\mathbb{R}^d} c(x, T(x)) \mu(dx) \longrightarrow \int_{\mathbb{R}^d \times \mathbb{R}^d} c(x, y) \pi(dx, dy)$$

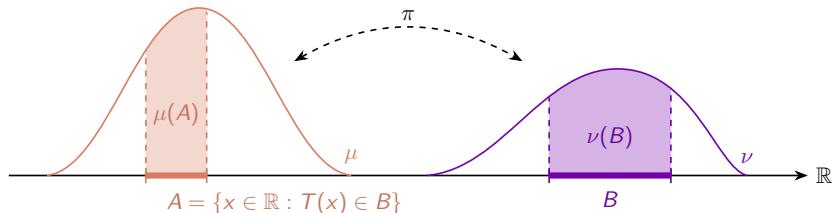
Quick reminders on Optimal Transport 2/3



► p -Wasserstein distance:

$$W_p(\mu, \nu)^p = \inf_{\pi \in \Pi(\mu, \nu)} \int d(x, y)^p \gamma(dx, dy)$$

Quick reminders on Optimal Transport 2/3



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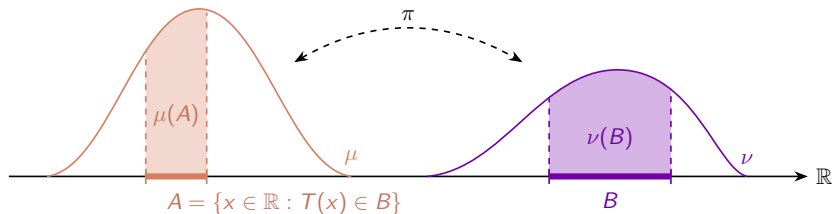
$$W_p(\mu, \nu)^p = \inf_{\pi \in \Pi(\mu, \nu)} \int d(x, y)^p \gamma(dx, dy)$$

- Kantorovich duality

$$\inf_{\pi \in \Pi(\mu, \nu)} \int c(x, y) \gamma(dx, dy) = \sup_{\phi} \int \phi(x) \mu(dx) + \int \phi^c(y) \nu(dy),$$

where $\phi^c(y) := \inf_x c(x, y) - \phi(x)$.

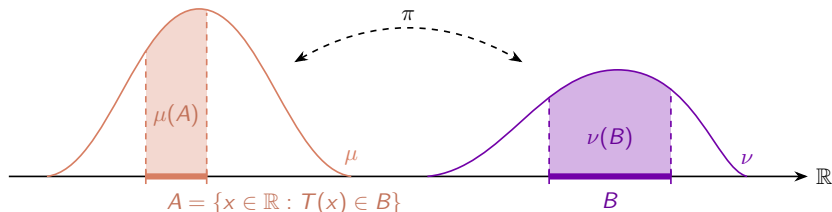
Quick reminders on Optimal Transport 3/3



► Kantorovich duality

$$T : \inf_{T_{\#}\mu=\nu} \int_{\mathbb{R}^d} c(x, T(x)) \mu(dx) = \sup_{\phi} \int \phi(x) \mu(dx) + \int \phi^c(y) \nu(dy)$$

Quick reminders on Optimal Transport 3/3



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► If an optimal transport map T^* exists, there also exists an optimal Kantorovich potential ϕ^* and

$$y = T^*(x) = \nabla \left(\frac{\|\cdot\|^2}{2} - \phi^* \right) (x) = x - \nabla \phi^*(x),$$

hence $-\nabla \phi^*(x) = T^*(x) - x = y - x$ is the **displacement** of the particle x .

What is the score function?

- **Forward process:** Ornstein–Uhlenbeck (OU) process:

$$d\vec{X}_t = -\vec{X}_t dt + \sqrt{2}d\omega_t, \quad \vec{X}_0 \sim \mu, \quad t \in [0, T].$$

Let $\vec{\rho}_t := \text{Law}(\vec{X}_t)$ and $\gamma \propto e^{-\|\cdot\|^2/2}$. Then $\vec{\rho}_t$ solves the Fokker–Planck equation

$$\partial_t \vec{\rho}_t(x) = \Delta \vec{\rho}_t(x) + \text{div}(\vec{\rho}_t(x)x).$$

whose stationary solution is γ . The process $(\vec{X}_t)_t$ is a **diffusive** process.

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- **Backward process:** $\tilde{X}_s := \vec{X}_{T-s}$ for $s \in [0, T]$. If $(\vec{X}_t)_{t \in [0, T]}$ is an OU process, then its time reversal $(\tilde{X}_s)_{s \in [0, T]}$ solves

$$d\tilde{X}_s = \left[\tilde{X}_s + 2 \underbrace{\nabla \ln(\vec{\rho}_{T-s}(\tilde{X}_s))}_{\text{The score function}} \right] ds + \sqrt{2}d\omega_s, \quad \tilde{X}_0 \sim \vec{\rho}_T.$$

Using OT to learn the score 1/2

- ▶ The Fokker–Planck equation is the 2-Wasserstein Gradient Flow (WGF) of the relative entropy $\mathcal{H}(\rho|\gamma) := \int V\rho + \int \rho \ln(\rho)$. That is:

$$\partial_t \rho_t(x) = \Delta \rho_t(x) + \operatorname{div}(\rho_t(x)x) = -\nabla \cdot (\rho_t(x)v_t(x)),$$

where $v_t(x) = -x - \nabla \ln(\rho_t(x))$.

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- ▶ Given a time-step $\tau > 0$, such a gradient flow can be discretized in time using the **JKO scheme**

$$\begin{cases} \rho_0 = \mu \\ \rho_{n+1}^\tau := \arg \min \mathcal{H}(\rho|\gamma) + \frac{1}{2\tau} W_2(\rho_n^\tau, \rho)^2 \end{cases}$$

Using OT to learn the score 2/2

► Solving

$$\inf_{\rho} \mathcal{H}(\rho|\gamma) + \frac{1}{2\tau} W_2(\rho_n^\tau, \rho)^2$$

amounts to finding the optimal transport map between

$$\rho_n^\tau \quad \text{and} \quad \rho_{n+1}^\tau \propto \exp\left(-\phi^{c/2\tau}\right) d\gamma,$$

where ϕ solves

$$\inf_{\phi} \int e^{-\phi^{c/2\tau}} d\gamma - \int \phi d\rho_n^\tau.$$

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► The optimal potential $\phi^\star$ satisfies

$$-\nabla \phi^\star(x) = \nabla(\phi^\star)^{c/2\tau}(y) = y - x.$$

Using OT to learn the score 2/2

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- The optimal potential ϕ^* satisfies

$$-\nabla\phi^*(x) = \nabla(\phi^*)^{c/2\tau}(y) = y - x.$$

- The displacement of the particle x being $v_t(x)$, we obtain

$$-\nabla\phi^*(x) = v_t(x) = -x - \nabla\ln(\rho_{n+1}^\tau(x)) \iff \boxed{\nabla\ln(\rho_{n+1}^\tau(x)) = x + \nabla\phi^*(x)}.$$